

Quiz 2

This is an individual assignment.

For the analytical problems below, you can:

1. write your answers in markdown cells with *L^AT_EX*, *or*
 2. Embed pictures with your handwritten answers inside the Notebook, *or*
 3. ``push`` a single pdf to your repository with all handwritten answers.
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Exercise 1 *3points*

Suppose you have a training set with N data points $\{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}$. Assume the samples are independent and identically distributed *i. i. d.*, and each sample is drawn from a Gaussian random variable with *known* mean μ and *unknown* variance σ^2 . The probability density function is defined as:

$$p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

where $\mu, \sigma > 0$.

Moreover, consider the Inverse Gamma as the prior probability on the hyperparameter σ^2 ,

$$p(\sigma^2|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right)$$

where $\alpha, \beta > 0$.

Answer the following questions:

1. 1point **Derive the maximum likelihood estimate MLE for the parameter σ^2 . Show your work.**

Ex1.

$$MLE: \mathcal{L}(\sigma^2) = \prod_{i=1}^N p(x_i | \mu, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(\frac{(x_i - \mu)^2}{-2\sigma^2}\right)}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \cdot e^{\left(\frac{\sum_{i=1}^N (x_i - \mu)^2}{-2\sigma^2}\right)}$$

$$\log_e \mathcal{L}(\sigma^2) = N \log_e \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \frac{1}{-2\sigma^2} \left(\sum_{i=1}^N (x_i - \mu)^2\right)$$

$$= -N \cdot \frac{1}{2} \left(\log_e(2\pi) + \log_e(\sigma^2)\right) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^N (x_i - \mu)^2\right)$$

$$\frac{\partial \log_e \mathcal{L}(\sigma^2)}{\partial \sigma^2} = 0 = -\frac{N}{2} \cdot \frac{1}{\sigma^2} - \frac{-1}{2(\sigma^2)^2} \left(\sum_{i=1}^N (x_i - \mu)^2\right)$$

$$\Rightarrow \frac{N}{2\sigma^2} = \frac{1}{2(\sigma^2)^2} \left(\sum_{i=1}^N (x_i - \mu)^2\right) \Rightarrow \sigma^2 = \frac{1}{N} \left(\sum_{i=1}^N (x_i - \mu)^2\right)$$

2. 1point **Derive the maximum a posteriori MAP estimate for the parameter σ^2 . Show your work.**

MAP: $p(\sigma^2|X) \propto p(X|\sigma^2) \cdot p(\sigma^2) \Rightarrow \log_e p(\sigma^2|X) = \log_e p(X|\sigma^2) + \log_e p(\sigma^2) + \text{constant}$

$$\log p(\sigma^2) = \alpha \log \beta - \log \Gamma(\alpha) - (\alpha+1) \log(\sigma^2) - \frac{\beta}{\sigma^2}, \log_e p(X|\sigma^2) = \frac{-N}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2 + \text{const}$$

$$\log_e p(\sigma^2|X) = \frac{-N}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2 - (\alpha+1) \log(\sigma^2) - \frac{\beta}{\sigma^2} + \text{const}$$

$$= \left(\frac{-N-2\alpha-2}{2} \right) \log(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^N (x_i - \mu)^2 + 2\beta \right) + \text{const}$$

$$\frac{\partial \log_e p(\sigma^2|X)}{\partial \sigma^2} = \left(\frac{-N-2\alpha-2}{2\sigma^2} \right) + \frac{1}{2(\sigma^2)^2} \left(\sum_{i=1}^N (x_i - \mu)^2 + 2\beta \right) = 0$$

$$\Rightarrow \frac{N+2\alpha+2}{2\sigma^2} = \frac{1}{2\sigma^2 \sigma^2} \left(\sum_{i=1}^N (x_i - \mu)^2 + 2\beta \right)$$

$$\Rightarrow \sigma_{MAP}^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2 + 2\beta}{N+2\alpha+2}$$

3. 1point **Show that the Inverse Gamma forms a conjugate prior with the Gaussian likelihood, and write down the pseudo-code for the online update of the prior parameters.**

Solution:

To show that Inverse Gamma is a conjugate prior, we need to demonstrate that the posterior has the same functional form as the prior.

From part 2, the log-posterior was:

$$\log p(\sigma^2|X) = - \left(\frac{N}{2} + \alpha + 1 \right) \log(\sigma^2) - \frac{1}{\sigma^2} \left(\frac{1}{2} \sum_{i=1}^N (x_i - \mu)^2 + \beta \right) + \text{const}$$

This can be rewritten as:

$$p(\sigma^2|X) \propto (\sigma^2)^{-(\alpha'+1)} \exp \left(-\frac{\beta'}{\sigma^2} \right)$$

where:

$$- \alpha' = \alpha + \frac{N}{2}$$

$$- \beta' = \beta + \frac{1}{2} \sum_{i=1}^N (x_i - \mu)^2$$

This is exactly the form of an Inverse Gamma distribution with updated parameters (α', β') . Therefore, Inverse Gamma is indeed a conjugate prior for the Gaussian likelihood with unknown variance.

Online Update Pseudocode:

...

Initialize: α_0, β_0 # prior parameters

For each new data point x_i :

Update parameters

$$\alpha_1 = \alpha_0 + 1/2$$

$$\beta_1 = \beta_0 + x_i - \mu^2/2$$

Updated parameters become new prior

$$\alpha_0 = \alpha_1$$

$$\beta_0 = \beta_1$$

Current MAP estimate

$$\sigma^2_{\text{MAP}} = \beta_1 / \alpha_1 + 1$$

...

The key insight is that each new observation updates the prior parameters incrementally, allowing for sequential Bayesian inference without storing all previous data points.

Exercise 2 3points

Motivation:

> Waste minimization is a critical aspect of sustainable production practices, particularly in the manufacturing of consumer goods. As the global demand for products continues to rise, so does the generation of waste, posing significant environmental and economic challenges. The production of consumer goods involves numerous stages, from raw material extraction to manufacturing, packaging, and distribution. At each step, there is the potential for waste generation, leading to resource depletion, pollution, and increased costs. Addressing this issue and implementing effective waste minimization strategies is essential to mitigate the environmental impact, conserve resources, and optimize production processes, all while maintaining profitability and meeting consumer expectations.

Suppose your company is in the business of manufacturing consumer goods products and that you are interested in minimizing waste production. Instead of producing a large number of items at any given time, design a strategy to determine how many products should be manufactured at any given time to minimize waste production.

1. 1point **The dataset $X = \{x_i\}_{i=1}^N$ is comprised of N samples x_i which represent the "number of products to be produced in a given interval of time". Which random variable RV best describes this data *assume x_i 's are i.i.d.*? For your choice of RV, write down the observed data likelihood, $\mathcal{L}$.

___ *Hint: select one from the most popular discrete RVs.* ___

Solution:

Since x_i represents the "number of products to be produced in a given interval of time", we are dealing with count data over fixed time intervals. The most appropriate discrete random variable for this scenario is the **Poisson distribution**.

The Poisson distribution is ideal because:

- It models the number of events *product production* occurring in a fixed interval
- It assumes events occur independently
- It's commonly used for manufacturing and production processes

For a Poisson distribution with parameter $\lambda > 0$:

$$p(x_i|\lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

The observed data likelihood for N i.i.d. samples is:

$$\begin{aligned}\mathcal{L}(\lambda) &= \prod_{i=1}^N p(x_i|\lambda) = \prod_{i=1}^N \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \\ &= \frac{\lambda^{\sum_{i=1}^N x_i} e^{-N\lambda}}{\prod_{i=1}^N x_i!}\end{aligned}$$

2. 1point **Solve for the parameter of the observed data likelihood using the MLE approach.**

Solution:

From part 1, the likelihood function is:

$$\mathcal{L}(\lambda) = \frac{\lambda^{\sum_{i=1}^N x_i} e^{-N\lambda}}{\prod_{i=1}^N x_i!}$$

Taking the log-likelihood:

$$\begin{aligned}\ell(\lambda) &= \log \mathcal{L}(\lambda) = \sum_{i=1}^N x_i \log \lambda - N\lambda - \log \left(\prod_{i=1}^N x_i! \right) \\ &= \left(\sum_{i=1}^N x_i \right) \log \lambda - N\lambda + \text{const}\end{aligned}$$

To find the MLE, we take the derivative with respect to λ and set it to zero:

$$\frac{\partial \ell}{\partial \lambda} = \frac{\sum_{i=1}^N x_i}{\lambda} - N = 0$$

Solving for λ :

$$\frac{\sum_{i=1}^N x_i}{\lambda} = N$$

$$\lambda = \frac{\sum_{i=1}^N x_i}{N}$$

Therefore, the MLE estimate is:

$$\hat{\lambda}_{MLE} = \frac{1}{N} \sum_{i=1}^N x_i = \bar{x}$$

The MLE for the Poisson parameter is simply the sample mean, which makes intuitive sense as λ represents the average rate of production.

3. *1point* ****Introduce a prior probability for the parameter of the data likelihood defined in 1 such that it forms a conjugate prior relationship, and justify your choice. Derive the solution for the parameter using the MAP approach.****

Solution:

For the Poisson likelihood, the ****Gamma distribution**** forms a conjugate prior for the parameter λ .

****Justification:**** The Gamma distribution is the natural conjugate prior for the Poisson likelihood because:

- Both involve exponential terms with λ

- The posterior will maintain the same functional form as the prior
- It's mathematically tractable and widely used in Bayesian analysis

The Gamma prior is defined as:

$$p(\lambda|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

where $\alpha > 0$ *shapeparameter* and $\beta > 0$ *rateparameter*.

MAP Derivation:

Using Bayes' theorem, the posterior is:

$$p(\lambda|X) \propto p(X|\lambda) \cdot p(\lambda)$$

From parts 1 and 2:

$$p(X|\lambda) \propto \lambda^{\sum_{i=1}^N x_i} e^{-N\lambda}$$

Combined with the Gamma prior:

$$p(\lambda|X) \propto \lambda^{\sum_{i=1}^N x_i} e^{-N\lambda} \cdot \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$= \lambda^{\sum_{i=1}^N x_i + \alpha - 1} e^{-(\beta + N)\lambda}$$

This is a Gamma distribution with updated parameters:

$$- \alpha' = \alpha + \sum_{i=1}^N x_i$$

$$- \beta' = \beta + N$$

For a Gamma distribution, the mode $MAP_{estimate}$ is:

$$\hat{\lambda}_{MAP} = \frac{\alpha' - 1}{\beta'} = \frac{\alpha + \sum_{i=1}^N x_i - 1}{\beta + N}$$

Exercise 3 *3points*

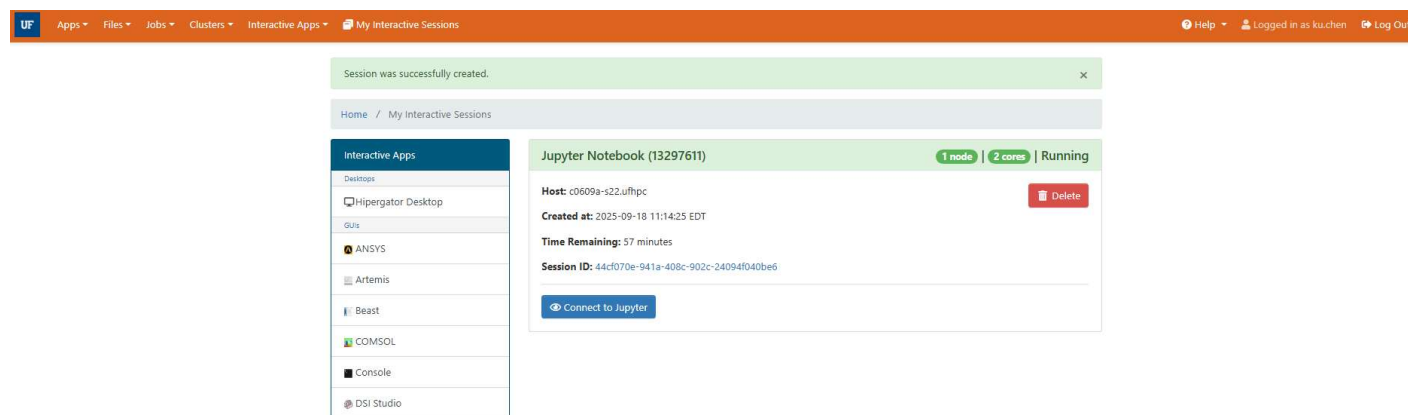
In this question, you will practice how to use HiPerGator and Git and GitHub to maintain your code.

1. *0.5points* **Open "Open On-Demand"

ood.rc.ufl.edu

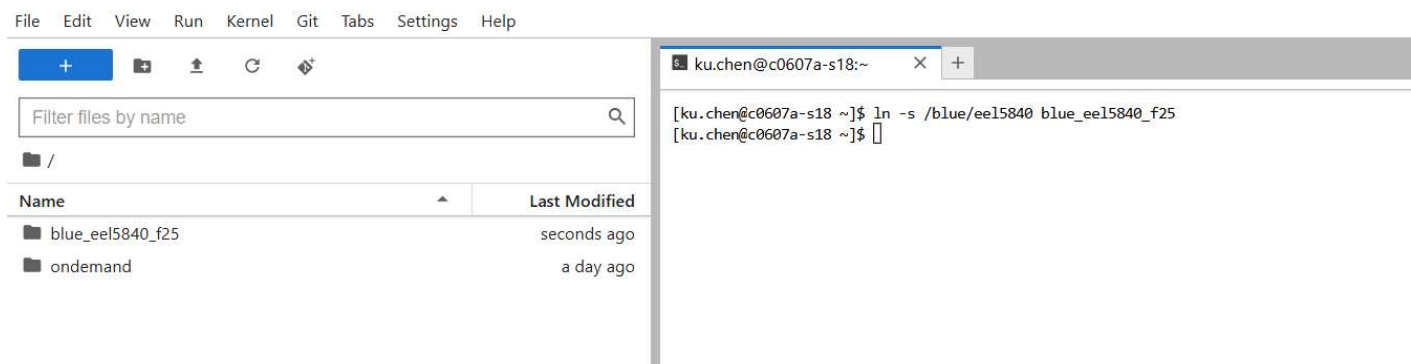
ood.rc.ufl.edu and create an interactive jupyter session with the following specifications: 2 CPU, 2 GPU and 16 GB of RAM. Attach a screenshot of your jupyter notebook card, your gatorlink should be visible *examplebelow*.**

```
from IPython.display import Image
Image('figures/ood_job_request.png', width=900)
```



2. *0.5points* **Create a symbolic link to map the class blue directory in your homepage. Attach a screenshot to show that this link has been created. Example and instructions below.**

```
Image('figures/symbolic_link.png', width=900)
```



3. *1point* **Using a terminal shell, change the directory to the `_eel5840_` group. Using my naming convention, you would run `cd blue_eel5840_f25/`. Next, if you don't see a folder with your GatorLink username, go ahead and create one `mkdir folder_name`. Finally make sure your folder is only accessible to you by changing its mode to 700 `chmod 700 folder_name`. Attach a screenshot with these steps. See example and instructions below.

Image('figures/chmod700.png', width=900)

Filter files by name

/ blue_eel5840_f25 /

Name	Last Modified
asihan	4 hours ago
blue_eel5840	3 days ago
blue_eel58...	2 days ago
eel5840	7 days ago
joung.dong...	17 hours ago
ku.chen	10 minutes ago
quiz-2-aliez...	6 days ago
share	23 days ago
Untitled.ipynb	2 hours ago

```

drwxrws--- 2 catiaspsilva eel5840 4096 Aug 27 16:33 share
drwx--S--- 2 shovon.m eel5840 4096 Sep 6 02:55 shovon.m
drwx--S--- 2 snangia eel5840 4096 Sep 6 02:56 snangia
drwx--S--- 3 sosorio2 eel5840 4096 Sep 16 22:46 sosorio2
drwx--S--- 2 s.swapnil eel5840 4096 Sep 6 02:58 s.swapnil
drwx--S--- 3 subashsatbhaskar eel5840 4096 Sep 18 02:49 subashsatbhaskar
drwx--S--- 3 sudipregmi eel5840 4096 Sep 17 20:21 sudipregmi
drwx--S--- 2 sujeongjo eel5840 4096 Sep 6 03:01 sujeongjo
drwx--S--- 2 sweller1 eel5840 4096 Sep 6 03:02 sweller1
drwx--S--- 3 tannerstaggs eel5840 4096 Sep 18 22:30 tannerstaggs
drwx--S--- 2 tboudreau1 eel5840 4096 Sep 6 03:04 tboudreau1
drwx--S--- 2 thadavale eel5840 4096 Sep 6 03:05 thadavale
drwx--S--- 3 thomas.martin eel5840 4096 Sep 18 22:47 thomas.martin
drwx--S--- 2 tochukwu.ogri eel5840 4096 Sep 6 03:07 tochukwu.ogri
drwx--S--- 2 trangkhong eel5840 4096 Sep 6 03:08 trangkhong
drwx--S--- 3 trevor.turnquist eel5840 4096 Sep 19 13:13 trevor.turnquist
drwx--S--- 5 tujuncheng eel5840 4096 Sep 18 15:51 tujuncheng
drwx--S--- 2 tyler.mark eel5840 4096 Sep 16 10:58 tyler.mark
-rw-r--r-- 1 mufansun eel5840 72 Sep 19 14:29 Untitled.ipynb
drwx--S--- 2 w.acostacruz eel5840 4096 Sep 6 03:10 w.acostacruz
drwx--S--- 2 will.english eel5840 4096 Sep 16 11:24 will.english
drwx--S--- 2 wjones2 eel5840 4096 Sep 6 03:11 wjones2
drwx--S--- 3 w.son eel5840 4096 Sep 13 20:53 w.son
drwx--S--- 2 wuy2 eel5840 4096 Sep 6 03:12 wuy2
drwx--S--- 2 xinyangwang eel5840 4096 Sep 6 03:13 xinyangwang
drwx--S--- 2 yanru.li eel5840 4096 Sep 6 03:14 yanru.li
drwx--S--- 3 jo.yeonggwon eel5840 4096 Sep 16 16:36 YeongGwon
drwx--S--- 3 y.ghadge eel5840 4096 Sep 19 12:11 y.ghadge
drwx--S--- 4 yipengliu eel5840 4096 Sep 16 16:06 yipengliu
drwx--S--- 5 yuyaolai eel5840 4096 Sep 17 11:43 yuyaolai
drwx--S--- 2 zachary.jenkins eel5840 4096 Sep 6 03:17 zachary.jenkins
drwx--S--- 2 zelinyang eel5840 4096 Sep 6 03:18 zelinyang
drwx--S--- 3 zelinyang eel5840 4096 Sep 16 21:59 zelinyang_task
drwx--S--- 3 zhangzihao eel5840 4096 Sep 16 11:22 zhangzihao
drwx--S--- 2 zhul1 eel5840 4096 Sep 6 03:19 zhul1
drwx--S--- 2 zixuan.liu eel5840 4096 Sep 19 08:58 zixuan.liu
[ku.chen@c0607a-s18 blue_eel5840_f25]$ ls -ld ku.chen
drwx--S--- 4 ku.chen eel5840 4096 Sep 19 15:55 ku.chen

```

4. 1point **In the terminal change the directory to your folder `\\cd folder_name/\\`, for me it's `cd catiaspsilva/`. This is where you will clone ALL assignments. Set up your GitHub SSH key, and clone your quiz 2 repository. Attach a screenshot with this step. See example and instructions below for cloning the repository.**

For instructions on setting up your GitHub SSH key, enroll in

thisworkshop

<https://ufl.instructure.com/enroll/TWR9LR>, select Module 3, and watch the 11-min video titled "Set up SSH Key" for a step-by-step hands-on demonstration.

Image('figures/cloning_repo.png', width=900)

Filter files by name 	
/ blue_eel5840_f25 / ku.chen /	
Name	Last Modified
 quiz-1-Ckci...	4 minutes ago

```
[ku.chen@c0607a-s18 ku.chen]$ git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
Cloning into 'quiz-1-Ckcinnabar'...
Username for 'https://github.com': ckcinnabar
Password for 'https://ckcinnabar@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git/'
[ku.chen@c0607a-s18 ku.chen]$ git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
Cloning into 'quiz-1-Ckcinnabar'...
Username for 'https://github.com': Ckcinnabar
Password for 'https://Ckcinnabar@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git/'
[ku.chen@c0607a-s18 ku.chen]$ git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
Cloning into 'quiz-1-Ckcinnabar'...
Username for 'https://github.com': Ckcinnabar
Password for 'https://Ckcinnabar@github.com':
[1]+  Stopped                  git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
[ku.chen@c0607a-s18 ku.chen]$ git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
fatal: destination path 'quiz-1-Ckcinnabar' already exists and is not an empty directory.
[ku.chen@c0607a-s18 ku.chen]$ Ckcinnabar
bash: Ckcinnabar: command not found
[ku.chen@c0607a-s18 ku.chen]$ git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
fatal: destination path 'quiz-1-Ckcinnabar' already exists and is not an empty directory.
[ku.chen@c0607a-s18 ku.chen]$ git clone https://github.com/UF-EEL5840-EEE4773-Fall-2025/quiz-1-Ckcinnabar.git
Cloning into 'quiz-1-Ckcinnabar'...
Username for 'https://github.com': Ckcinnabar
Password for 'https://Ckcinnabar@github.com':
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 15 (delta 3), reused 2 (delta 2), pack-reused 9 (from 1)
Receiving objects: 100% (15/15), 1.48 MiB | 11.22 MiB/s, done.
Resolving deltas: 100% (4/4), done.
[ku.chen@c0607a-s18 ku.chen]$ ls
quiz-1-Ckcinnabar  quiz-1-Ckcinnabar5
[ku.chen@c0607a-s18 ku.chen]$ rm -rf quiz-1-Ckcinnabar5
[ku.chen@c0607a-s18 ku.chen]$ ls
quiz-1-Ckcinnabar
```

On-Time *1point*

Submit your assignment before the deadline.

Submit Your Solution

Confirm that you've successfully completed the assignment.

Along with the Notebook, include a PDF of the notebook with your solutions.

``add`` and ``commit`` the final version of your work, and ``push`` your code to your GitHub repository.

Submit the URL of your GitHub Repository as your assignment submission on Canvas.

